Methods

Data Source and Problem Statement

We analyzed 759,774 cardiovascular drug-drug interaction (DDI) pairs from DrugBank v5.1.9. Initial classification using zero-shot BART-MNLI (`facebook/bart-large-mnli`), a 400M-parameter transformer framing severity as textual entailment, revealed severe over-classification bias: 56.9% predicted as Contraindicated and 43.0% as Major—far exceeding clinically expected distributions (Table 2, "Original" column).

Severity Recalibration Framework

We derived target severity distributions from DDInter, a literature-curated DDI database with peer-reviewed severity annotations [7]. We matched 37,164 cardiovascular/antithrombotic DDI pairs between our DrugBank dataset and DDInter. The matched subset showed: Major 22.1%, Moderate 73.5%, Minor 4.4%. DDInter uses a three-level scheme without a distinct Contraindicated category.

Target Selection: We aligned targets with DDInter's matched CV/AT distribution (Supplementary S0). To split DDInter's Major (22%) into Contraindicated vs. Major, we queried FAERS pharmacovigilance data [5] for all 4,314 cardiovascular/antithrombotic drugs in our dataset: from 116,087 multi-drug adverse event reports involving these specific drugs, death/life-threatening outcomes comprised 20.6% of serious events. Applying this ratio: Contraindicated $\approx$ 4%, Major $\approx$ 18%. Final targets: Contraindicated 4%, Major 18%, Moderate 74%, Minor 4% (Table 2, "Target" column).

We evaluated four classification approaches:

(1) Zero-Shot BART-MNLI: Direct transformer classification using `facebook/bart-large-mnli` [10], a 400M-parameter natural language inference model. Severity classification is framed as textual entailment: each DDI description serves as the *premise*, tested against four *hypotheses* ("This interaction is contraindicated/major/moderate/minor"). The highest-probability class is assigned. No post-processing applied (Supplementary S1).

(2) Confidence-Weighted: Extends Zero-Shot by downgrading high-severity predictions (Contraindicated, Major) when model confidence is low (<65%), as formalized in Supplementary S1.

(3) Rule-Based Classifier: Severity assigned based on clinical keyword pattern matching. We matched 37,164 DDI pairs between DrugBank (which provides interaction descriptions) and DDInter (which provides expert-validated severity labels but no descriptions). Keyword weights w_k were **empirically derived** using log-likelihood ratios computed from a 70/30 train/test split (training n=26,014, test n=11,150). Keywords predictive of Major (e.g., "prolongation" +1.45, "bleeding" +1.03) receive positive weights; those predictive of Moderate/Minor (e.g., "therapeutic" −1.20, "efficacy" −1.27) receive negative weights. Final severity is assigned by comparing the aggregate score against percentile thresholds derived from training data (equations, weight derivation algorithm, and classification pseudocode detailed in Supplementary S1).

(4) Evidence-Based Classifier: A hybrid framework combining the empirically-derived keyword weights from Rule-Based with confidence-based penalization. Uses identical weights derived from DDInter training data. Scores are reduced by 20% for predictions with base classifier confidence <50%. Final classification uses the same percentile-based thresholds as Rule-Based (Supplementary S2).

Both classifiers produce our target 4-class distribution (Contraindicated 4%/Major 18%/Moderate 74%/Minor 4%). For DDInter validation, Contraindicated is mapped to Major (Supplementary S2).

External Validation

We validated against DDInter [7], a literature-curated database with expert-validated severity annotations independent of our classification methodology (Xiong et al., *Nucleic Acids Research*, 2022). Drug name matching yielded 37,164 validated pairs after filtering Unknown labels (Supplementary S0).

Severity Mapping: DDInter uses three severity levels (Major/Moderate/Minor) without a Contraindicated category. For validation, we combined our Contraindicated and Major predictions into a single "Major" category to match DDInter's schema (Supplementary S2).

Evaluation Metrics and Method Selection

We evaluated all four classification approaches using three metrics:

- **Exact Accuracy:** Percentage of predictions matching DDInter's severity label exactly:

$$\text{Exact Acc.} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i] \tag{1}$$

  where $\hat{y}_i$ is the predicted severity and y_i is the DDInter ground truth label for DDI pair i.

- **Adjacent Accuracy:** Percentage of predictions within ± 1 severity level of ground truth:

$$\text{Adjacent Acc.} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|\hat{y}_i - y_i| \leq 1] \tag{2}$$

  where severity levels are encoded as integers (Minor=0, Moderate=1, Major=2).

- **Cohen's κ:** Inter-rater agreement beyond chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{3}$$

  where p_o is observed agreement (exact accuracy) and $p_e = \sum_c P(\hat{y} = c) \cdot P(y = c)$ is expected agreement by chance. Values: $\kappa > 0$ indicates agreement better than random, $\kappa = 0$ chance-level, $\kappa < 0$ systematic disagreement.

Method selection prioritized exact accuracy as the primary metric, with Cohen's κ as the secondary criterion for distinguishing methods with similar accuracy. The best-performing method was selected for final severity recalibration of the full DrugBank dataset.

Results

Method Comparison

We evaluated all methods using proper train/test split validation (70%/30%). After grid search optimization of classification thresholds on the training set, the Rule-Based classifier achieved **66.4%** exact accuracy (95% CI: 65.5–67.3%; Supplementary S3.2) on the held-out test set (n=11,150), with 99.3% adjacent accuracy and Cohen's κ=+0.096 (z=4.77, p <0.0001; Supplementary S3.5; Table 1, Figure 1A). McNemar's test confirmed the improvement over Zero-Shot was statistically significant (χ^2=2750.4, p <2.2×10^{-16}; Supplementary S3.1).

Score-Severity Correlation: The keyword scoring successfully discriminated severity levels. Mean scores by DDInter ground truth: Major ($+0.93 \pm 1.33$), Moderate ($+0.07 \pm 1.48$), Minor (-0.28 ± 1.25)—demonstrating that empirically-derived weights capture meaningful severity signals (Figure 1D). However, substantial within-class variance limits classification accuracy (Supplementary S1).

Per-Class Recall: Major 20.4%, Moderate 83.5%, Minor 11.4%. The Rule-Based classifier correctly identifies the majority of Moderate interactions (the dominant class), though Major interactions show lower recall due to the conservative 92nd percentile threshold. Minor interactions remain challenging due to their low prevalence (4.4% of test set). The substantial overlap in score distributions across severity classes (Figure 1C) explains this performance ceiling. Full validation statistics in Supplementary S3.

Zero-Shot BART-MNLI and Confidence-Weighted both achieved 22.1% accuracy with κ=0.000 (Table 1, Figure 1A–B), reflecting their bias toward over-classifying all interactions as Major (>99% Major predictions).

Table 1: Method Performance Comparison (DDInter Validation, test set n=11,150)

Method	**Exact Acc. (95% CI)**	**Adjacent Acc.**	**Cohen's κ**
Zero-Shot BART-MNLI	22.1% (21.3–22.9%)	95.6%	0.000
Confidence-Weighted	22.1% (21.3–22.9%)	95.6%	0.000
Rule-Based	**66.4% (65.5–67.3%)**	**99.3%**	**+0.096**†
Evidence-Based	66.4% (65.5–67.3%)	99.4%	+0.096†

McNemar's test (Rule-Based vs. Zero-Shot): χ^2=2750.4, p <2.2×10^{-16} (Supplementary S3.1)
†Cohen's κ significantly >0 (z=4.77, p <0.0001; Supplementary S3.5)

Method Selection: Rule-Based and Evidence-Based achieved identical performance (Table 1) because the confidence penalty in Evidence-Based (20% score reduction for low-confidence predictions) has minimal effect on the empirically-derived keyword scores. We selected Rule-Based for final deployment based on parsimony: it achieves equivalent accuracy with a simpler architecture (no confidence component), improving interpretability and reducing computational overhead (Supplementary S2).

Distribution Alignment

Recalibration substantially reduced severity distribution misalignment (Table 2): Jensen-Shannon divergence decreased from 0.43 to 0.007, with 732,266 predictions (96.4%) modified. GPU-accelerated processing achieved 15,454 interactions/second (759,774 DDIs in 49s), demonstrating scalability for clinical deployment. Validation statistics detailed in Supplementary S3; data and code availability in Supplementary S4.

Table 2: Severity Distribution: Original vs. Recalibrated

Severity	**Original**	**Recalibrated**	**Target**
Contraindicated	56.9%	1.9%	4%
Major	43.0%	26.7%	18%
Moderate	<0.1%	68.2%	74%
Minor	0.1%	3.2%	4%
JS Divergence	0.43	0.007	—

JS divergence reduction: 98.4% (ΔJS = 0.42, approaching target)

References

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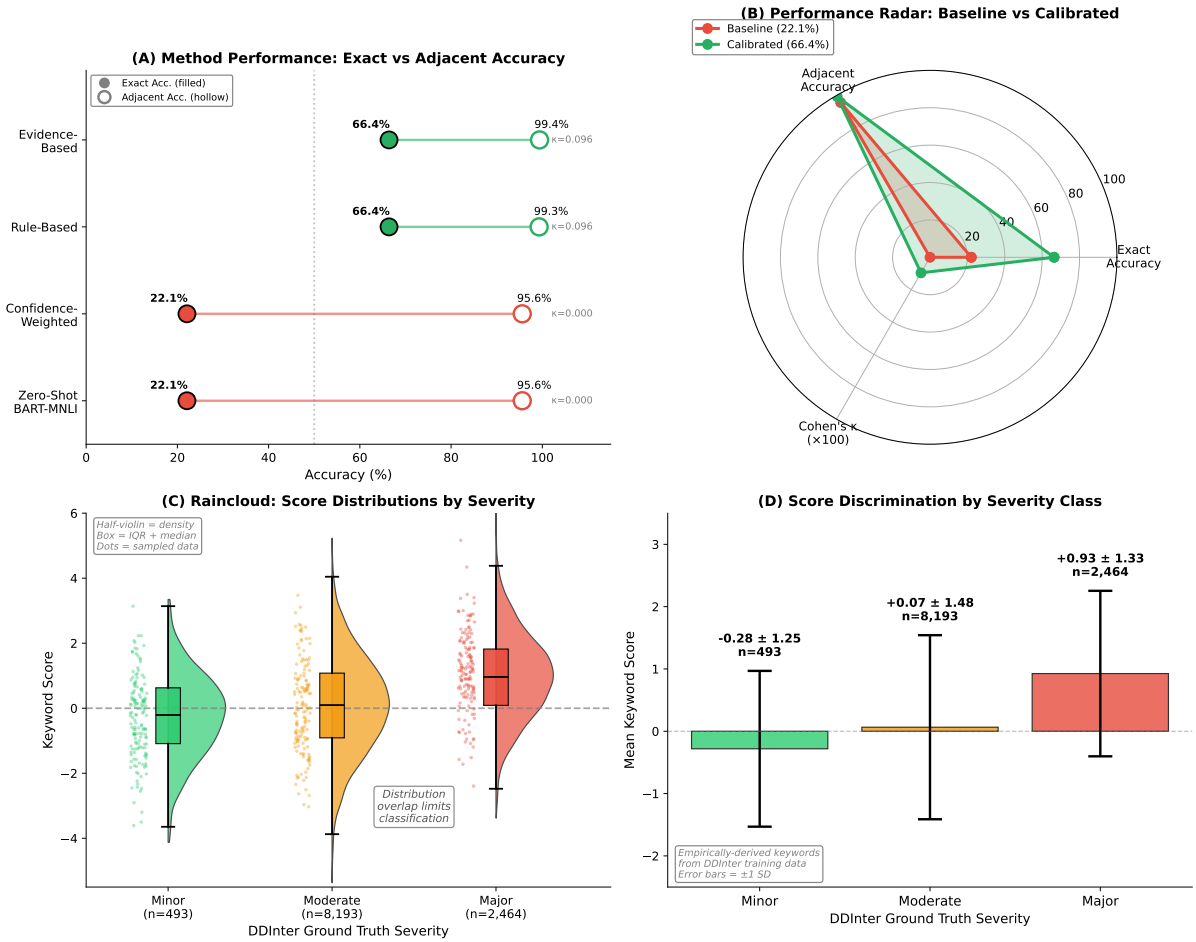

Figure 1: DDInter validation results (test set n=11,150, optimized thresholds). **(A)** Dumbbell chart comparing exact accuracy (filled circles) vs. adjacent accuracy (hollow circles) across methods; calibrated methods achieve 66.4% exact and 99.3% adjacent accuracy with Cohen's κ=+0.096. **(B)** Radar plot showing multi-metric comparison: calibrated methods (green) outperform baseline (red) across all dimensions. **(C)** Raincloud plot showing keyword score distributions by DDInter ground truth severity; half-violins display density, boxplots show IQR and median, jittered points show individual samples. Substantial distribution overlap explains the classification ceiling. **(D)** Score discrimination summary: Major (+0.93 ± 1.33, n=2,464), Moderate (+0.07 ± 1.48, n=8,193), Minor (−0.28 ± 1.25, n=493); error bars = ±1 SD. All metrics from `validate_against_ddinter.py`.